3rd Talk: Applications & Open Problems

Goal :- ① To describe applications of $\mathcal{E}(X)$, the group of (based) self homotopy equivalences of X.

② Direct to some open problems.

Applications

[A] Classification of Hurewicz fibration over circle.

Recall that ① $p: E \to B$ is a Hurewicz fibration with fibre F if for any top. space X, following com. diagram has a lift:

$$\begin{array}{ccc} X \times \{0\} & \longrightarrow & E \\ \downarrow & \nearrow & \downarrow p \\ X \times [0,1] & \longrightarrow & B \end{array}$$

that commutes and

$\exists$ a htpy equiv. $g : F \xrightarrow{\simeq} p^{-1}(*)$ for some $* \in B$.

② Two fibrations $p_i : E_i \to B$ with fibre F and htpy equiv. $g_i : F \to p_i^{-1}(*)$ are called <u>equivalent</u> if $\exists$ htpy equiv. $f : E_1 \to E_2$ such that $g_2 \simeq f \circ g_1$ and $p_1 \simeq p_2 \circ f$.

$$\begin{array}{ccccc} F & \xrightarrow{g_1} & E_1 & \xrightarrow{p_1} & B \\ \| & \textcircled{\simeq} & \downarrow f & \textcircled{\simeq} & \| \\ F & \xrightarrow{g_2} & E_2 & \xrightarrow{p_2} & B \end{array}$$

- This is an equivalence relation.

Define $\mathcal{H}(B, F)$ to be the collection of equivalence classes of fibrations over B with fibre F.

[Allaud '66] There is a fibration $p_\alpha : E_\alpha \to B_\alpha$ with htpy equiv. $g_\alpha : F \to p_\alpha^{-1}(*)$ s.t. there is a one-one correspondence:-

$$[B, B_\alpha]_* \longleftrightarrow \mathcal{H}(B, F).$$

where $[B, B_\alpha]_*$ is the collection of based htpy class of maps from B to B_α.

⊠ Also, $\pi_i(B_\alpha, *) \cong \pi_{i-1}\left(\mathcal{E}(F)_f, 1_F\right)$, where $\mathcal{E}(F)_f$ is the collection of all free self-equivalences of F.

↖ need not preserve base point

In particular, taking $B = S^1$, we see from the first corres. that $\mathcal{H}(S^1, F) \simeq [S^1, B_\alpha]_* = \pi_1(B_\alpha, *) \cong \pi_0\left(\mathcal{E}(F)_f, 1_F\right) \cong \mathcal{E}(F)_f$

Therefore, the equivalence class of fibrations over S^1 with fibre F corresponds to $\mathcal{E}(F)_f$, the group of htpy class of free self-equivalences of F.

Note that, if $\pi_1(F)$ is trivial or F is an H-space, then $\mathcal{E}(F)_f \cong \mathcal{E}(F)$

Thus knowing the group $\mathcal{E}(F)$ in specific cases gives us the characterization of fibrations over S^1 upto equiv.

[B] Action of $\mathcal{E}(X) \times \mathcal{E}(Y)$ on $[X, Y]$.

Define $(\mathcal{E}(X) \times \mathcal{E}(Y)) \times [X, Y] \longrightarrow [X, Y]$ by

$$((\alpha, \beta), \varphi) \longmapsto \beta \circ \varphi \circ \alpha^{-1}$$

- Check that this is a well-defined action.

similarly, $\mathcal{E}(X)$ acts on $[X\times X, X]$ as well, as follows -

$$\mathcal{E}(X) \times [X\times X, X] \longrightarrow [X\times X, X]$$
$$(\alpha, \varphi) \longmapsto \alpha \circ \varphi \circ (\alpha^{-1} \times \alpha^{-1})$$

Note that, an H-space structure on X is a map $M: X\times X \to X$ such that there exists $e \in X$, $M(e,-), M(-,e): X \to X$ are homotopic to Id_X.

Let $\mathcal{M}(X) := \{[M] : M \text{ is an H-space structure on } X\} \subseteq [X\times X, X]$

$\mathcal{M}_0(X) := \{[M] : M \text{ is a htpy associative H-space struc. on } X\}$
$\subseteq \mathcal{M}(X)$.

The action of $\mathcal{E}(X)$ on $[X\times X, X]$ induces an action on $\mathcal{M}(X)$ as well as on $\mathcal{M}_0(X)$.

Take $\tilde{M}(X) = M(X)/\Sigma(X)$ and $\tilde{M}_0(X) = M_0(X)/\Sigma(X)$

[Urjel '68] For a finite, Rtpy associative H-complex X, $\tilde{M}_0(X)$ is always finite, and $\tilde{M}(X)$ is infinite if and only if (the n-th Betti number number of $X \vee X$) $\times$ (rank of $\pi_n(X)$) $\neq 0$.

[C] Postnikov invariants :-

The above action also determines non-uniqueness of postnikov invariants.

[Adams '56] If X' is an n-section, $n \geq 2$ and G an abelian group, then the Rtpy types of (n+1)-sections with have n-type X' and (n+1)-th Rtpy group G is in 1-1 correspondence with

$$[X', K(G, n+2)]/\Sigma(X') \times \mathrm{Aut}(G) \qquad [\text{as } \Sigma(K(G,n+2)) = \mathrm{Aut}(G)]$$

- A space X' is called an <u>n-section</u> if $\pi_i(X') = 0$ for $i > n$.
- The <u>n-type</u> of X is the htpy type of any n-section X' s.t. $\exists$ a map $p: X \to X'$ with $p_*: \pi_i(X) \xrightarrow{\cong} \pi_i(X')$ for $i \leq n$

↳ one can always take X_n, the n-th space in Postnikov decom. of X, for X' as the n-type of X.

Thus from the above result, we have, the htpy type of X_{n+1}, the $(n+1)$-th Postnikov space of X, is uniquely determined by the htpy type of X_n, $(n+1)$-th htpy group $\pi_{n+1}(X)$, and the equiv. class k^{n+1} of the postnikov invariant in $\left[X_n, K(\pi_{n+1}(X), n+2)\right] \Big/ \Sigma(X_n) \times Aut(\pi_{n+1}(X))$

$H^{n+2}(X_n, \pi_{n+1}(X)) / \ldots$

Therefore for uniqueness of Postnikov invariants, one should really consider those invariants as element of $H^{n+2}(X_n, \pi_{n+1}(X)) / \Sigma(X_n) \times Aut(\pi_{n+1}(X))$

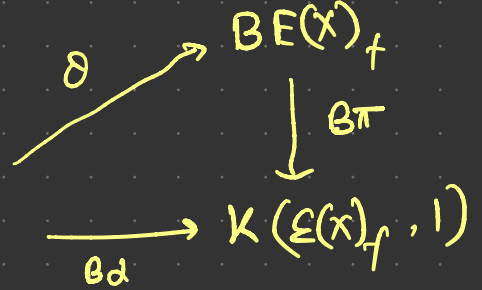
↳ this in particular determines obstruction classes for extending a map.

(0) Htpy actions of a group on a space :-

- A htpy action of a group G on a space X is a homomorphism
$$\alpha : G \to \mathcal{E}(X)_f$$
- If there exist such a map for a space X, then X is called htpy G-space.
- If for a htpy G-space X, the action map α factors through homeomorphism group, $\mathrm{Homeo}(X)$, of X, then X is called G-space. [i.e. X has an actual action of G]

Q: When a htpy G-action is realizable by a G-action?

[Cooke '78] A htpy G-action is realizable by a top. G-action iff $\exists$ a map $\theta : K(G,1) \to B\,E(X)_f$ s.t. following diagram is htpy commutative :-



Here $B\,E(X)_f$ is the classifying space for the monoid $E(X)_f$ via the Dold-Lashof construction.

$\alpha : G \to E(X)_f$ is the htpy G-action map.

$\pi : E(X)_f \to \mathcal{E}(X)_f$ is the quotient map (quotienting by htpy).

→ This induces map on respective classifying spaces but $BG \simeq K(G,1)$ and $B\mathcal{E}(X)_f \simeq K(\mathcal{E}(X)_f, 1)$. so the above multi

using above result, Zabrodsky '82 observed the following:-

Propⁿ:- If X is a CW-complex, then $\exists$ a space Y of same htpy type as X such that every htpy class in $\mathcal{E}(X)_f$ contains a homeomorphism.

Open Problems

Q.1 a) Characterize those finite, connected complex X for which $\mathcal{E}(X)$ is finitely generated / finitely presented.

The Sullivan-Wilkerson theorem asserts that for simply connected finite complex X (i.e. a finite CW complex or has finitely many non-zero htpy groups). $\mathcal{E}(X)$ is finitely presented.

But for general finite complex $\mathcal{E}(X)$ is infinitely generated.

one such example was given by [Frank-Kahn '77].

b) Is there a finite, connected X, with $\mathcal{E}(X)$ finitely generated but not finitely presented.

As there are many more finitely gen. groups than finitely presented groups. so somewhat interesting to know such example.

c) Determine spaces X for which $\mathcal{E}(X)$ is trivial.

In general, for non-contractible space X, $\mathcal{E}(X)$ trivial is very rare

one such case is $K(\mathbb{Z}_2, n)$. as $\mathcal{E}(K(\mathbb{Z}_2, n)) \cong Aut(\mathbb{Z}_2) = \{1\}$.

Kahn '76 also gave non-trivial example of spaces with trivial $\mathcal{E}(X)$.

so in a way, this is nice direction to pursue.

[a'] For simply connected finite CW-complex X, can we have the finite generation of the group $\mathcal{E}(\Omega X)$?

This is a question similar to (a) but for H-space ΩX, which is as a complex infinite one.

Also, a useful one in many places.

Q.2 Calculate $\mathcal{E}(X)$ for various well-known spaces, for example,

$X = \mathbb{H}P^n$ for $n > 2$

X = any rank-2 H-spaces.

X = compact Lie group, or Kähler manifold.

Q.3 For Moore spaces $M(G, n)$ $(n \geq 2)$, there is an epimorphism $\mathcal{E}(M(G,n)) \to \mathrm{Aut}(G)$ with kernel $\mathrm{Ext}(G, \pi_{n+1}(M(G,n)))$. Calculate the extension for $\mathcal{E}(M(G,n))$ precisely.